

Analysis — from a clean dataset to an explained change

Step 4 of the weekly workflow: four angles that turn 'the number moved' into 'this is why, and this is what it means'

1 Cohorts

Players grouped by registration or FTD date and compared at equal cohort age. This separates a genuine quality change from a change in traffic mix.

D1 / D7 / D30 curves
Equal cohort age
Registration vs FTD anchor

2 Segmentation

The same metric split by value tier, geo, device and acquisition channel. An average that hides two opposite movements is worse than no metric.

VIP / mid / low
Geo & device
Channel & partner

3 Funnel

Step-by-step conversion from landing to first deposit, so a drop can be located at a stage instead of being reported as a total.

Landing → form
Form → registration
Registration → FTD

4 Root cause

Before any business explanation, the boring causes are ruled out: tracking, promo calendar, seasonality, mix shift. Only then behaviour.

Tracking change?
Promo overlap?
Mix shift?

A worked example — “D7 retention dropped 4 points last week”

The same drop can have five different causes, and four of them are not a product problem. This is the order I eliminate them in.

1. Is it real?

Check the data first

Did a tracking tag change, did a source arrive late, did the cohort definition shift? Roughly half of sudden drops die here.

Result: no pipeline change found

2. Is it the mix?

Same players, new blend

A cheap channel scaled up last week. Per-channel retention is unchanged — only the weighting moved.

Result: partly, +60% traffic from one partner

3. Is it the promo?

Compare like with like

The prior week contained a reload bonus that kept players active on day 7. The baseline itself was inflated.

Result: yes, prior week had a campaign

4. Is it seasonal?

Calendar effects

Sports calendar, holidays and payday cycles in the main geos are checked against the same week last year.

Result: no material calendar effect

5. Is it product?

Only what remains

After the above, the residual drop is isolated to one segment — and that is the finding worth a recommendation.

Result: ~1.2 pts, mobile iOS cohort

Methods and the traps that produce confident nonsense

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The techniques I rely on for weekly analysis, and the mistakes that make a chart look convincing while being wrong

Methods used in a weekly cycle

- **Cohort retention curves**

Classic retention by registration cohort, always compared at equal age. Rolling retention reported separately, never mixed into the same chart.

- **Value segmentation**

Players banded by deposit and margin contribution. Most weekly movement in GGR comes from a small group, so the average alone explains nothing.

- **Funnel decomposition**

Conversion split by stage, then by channel and device, to locate the step where the loss actually happens.

- **Contribution analysis**

A total change broken into volume effect and rate effect: did we get fewer players, or did the same players convert worse?

- **Week-over-week with baselines**

Every comparison carries an expected range built from prior weeks, so normal volatility is not reported as a trend.

- **Anomaly triage**

Deviations sorted by business impact, not by percentage — a 40% move on a tiny segment rarely deserves attention first.

- **Player lifecycle view**

New, active, dormant and reactivated tracked as flows between states, which explains a flat total that hides heavy churn and heavy acquisition.

- **Bonus impact isolation**

Promotional and organic activity separated so campaign effects are measured, not absorbed into the trend.

- **Payback and LTV horizon**

Acquisition judged on payback period rather than first-week revenue, since a channel can look expensive and still be the most profitable one.

- **Game and provider mix**

Revenue split by game category and provider, which often explains a margin change that looks like a player-behaviour change.

- **Repeat-deposit behaviour**

Time from first to second deposit is an early signal that moves weeks before retention curves show anything.

- **Cross-check against a second source**

Any finding that would change a budget is reproduced from an independent path before it is presented.

Traps I actively check for

- **Simpson's paradox**

Every segment improves while the total gets worse, because the mix shifted. The reason I never report a total without at least one split.

- **Survivorship in cohorts**

Comparing a 30-day-old cohort with a 3-day-old one measures cohort age, not performance.

- **Correlation presented as cause**

Two lines moving together is a hypothesis, not a finding — I state it as one until something tests it.

- **Small-sample noise**

A 50-player segment swinging 20% is arithmetic, not behaviour. Minimum sample sizes are set before slicing.

- **Cherry-picked window**

Moving the start date until the trend appears. The comparison period is fixed in step 1, before results are seen.

- **Averages over skewed data**

iGaming revenue is extremely skewed — median and percentiles reported alongside the mean, or one VIP defines the week.

- **Slicing until something appears**

Enough segments guarantee a false positive somewhere. Splits are decided from the hypothesis, not discovered by scanning.

- **Ignoring the denominator**

A rising conversion rate on collapsing traffic is not good news; absolute and relative are always shown together.

What leaves this step

- The change, quantified: how much moved, in which segment, over which period.
- The cause, with alternatives already ruled out and named.
- Confidence level, and what would be needed to raise it.
- The one finding nobody asked about but everybody needs to know.
- Guardrail metrics checked: what did not get worse while the primary metric moved.
- Open questions this cycle could not answer, and what data would be needed next time.
- The exact filters and period behind every figure, so the analysis can be reproduced.

Findings go into the dashboard only once the explanation survives the checks — a fast wrong answer costs more than a slow right one.

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